

On the number of crossings of small bipartite graphs

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Abstract

We study rectilinear drawings of the bipartite graph $K_{3,n}$ under a *corner constraint*: all red vertices lie in the exterior corner determined by one blue vertex. We prove, by a short geometric pair-counting argument, that for any such drawing the total number of crossings satisfies

$$\binom{n}{2} \leq \text{cr} \leq 3\binom{n}{2}.$$

Moreover, we give an explicit placement achieving the lower bound $\text{cr} = \binom{n}{2}$, hence the bounds are optimal within this constrained family. The proof formalizes the T_1/T_2 case analysis implicit in the sketches: each unordered pair of red points contributes either 1 or 3 crossings in total, depending on the region in which the second point lies relative to the first.

1 Alternate way to count crossings in $K_{3,n}$

Introduction

We have three fixed blue points b_1, b_2, b_3 that form a triangle (no three points we use are collinear). We have n red points. From each red point we draw straight segments to all three blue points. We count only crossings between edges that do not share an endpoint.

A rule we will use

Opposite-sides rule. For four points A, B, C, D in general position, the segments AB and CD cross inside if and only if

- C and D are on opposite sides of the line AB , and
- A and B are on opposite sides of the line CD .

How one red pair behaves

Fix one red point p . Draw the three lines pb_1, pb_2, pb_3 . These three lines split the plane into six open regions (call them *wedges*). Take another red point $q \neq p$ and look at the three blue pairs $\{b_1, b_2\}, \{b_2, b_3\}, \{b_1, b_3\}$. For each blue pair, apply the opposite-sides rule to the two independent edges in the little $K_{2,2}$ on $\{b_i, b_j, p, q\}$.

Lemma 1 (Each wedge is fixed). *For fixed p , the YES/NO answers for the three blue pairs do not change. In fact, there are only two kinds of wedges:*

- in two wedges , all three blue pairs give a crossing;
- in the other four wedges, exactly one blue pair gives a crossing.

Proof. For a fixed blue pair $\{b_i, b_j\}$ the opposite-sides rule depends only on whether the direction of the line pq lies between the directions from p to b_i and b_j . If q does not cross any of the lines pb_1, pb_2, pb_3 , this position cannot change, so the YES/NO answer is constant inside each wedge. Checking one sample point in each wedge shows that two wedges produce YES for all three blue pairs, and the other four produce YES for exactly one blue pair. $\square$

Let

$$\tau(p, q) := \text{number of crossings contributed by the red pair } \{p, q\}.$$

By the lemma, $\tau(p, q) = 3$ when q is in one of the two “all-YES” wedges of p (we will call these *flank* wedges), and $\tau(p, q) = 1$ when q is in any of the other four *middle* wedges.

Lemma 2. *Every crossing in the whole drawing belongs to exactly one red pair $\{p, q\}$. Therefore*

$$\text{Total crossings} = \sum_{\{p, q\}} \tau(p, q).$$

Proof. A crossing always occurs inside one of the three little $K_{2,2}$ ’s made by a blue pair and a red pair. That crossing is counted in the $\tau(p, q)$ for that red pair and for no other pair. $\square$

The payoff

Let N_F be the number of red pairs that are *flank* (i.e., $\tau = 3$), and let $N_M = \binom{n}{2} - N_F$ be the number that are *middle* (i.e., $\tau = 1$).

Theorem 3. *For any placement of the red points,*

$$\text{Total crossings} = 3 N_F + 1 N_M = \binom{n}{2} + 2 N_F.$$

In particular,

$$\binom{n}{2} \leq \text{Total crossings} \leq 3 \binom{n}{2}.$$

Both ends are achievable:

- To get the minimum $\binom{n}{2}$, put all red points on a very thin line like cluster so that every pair of red points is in a middle wedge.
- To get the maximum $3\binom{n}{2}$, cluster all red points tightly inside a flank wedge so every pair is flank.

Setup

We have three blue points b_1, b_2, b_3 that form a triangle. We have n red points. For each red point r and each blue point b_i we draw the straight segment rb_i . We count only crossings that happen *inside* both segments and only for edges that *do not* share an endpoint. Assume general position (no three of our points are collinear).

Two-separators rule. For four points A, B, C, D , the segments AB and CD cross inside *iff*

- C and D are on **opposite sides** of the line AB , and
- A and B are on **opposite sides** of the line CD .

We will apply this with $AB = b_i b_j$ (a blue line) and $CD = pq$ (a red line).

Six wedges around one red point

Fix a red point p inside the triangle. Draw the three lines pb_1, pb_2, pb_3 . They split the plane into six open regions, which we call *wedges*. Pick another red point q . For each of the three blue pairs $\{b_1, b_2\}, \{b_2, b_3\}, \{b_1, b_3\}$, apply the two-separators rule to decide if the independent edges cross.

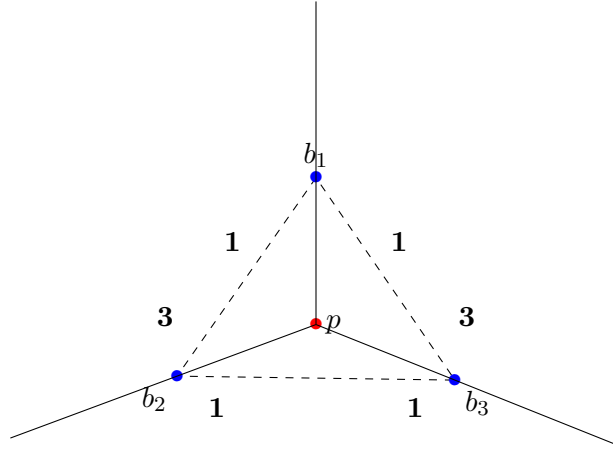


Figure 1: Six wedges around p cut by pb_1, pb_2, pb_3 . Two *flank* wedges are labeled 3; the four *middle* wedges are labeled 1.

Wedge-by-wedge proofs

For a wedge W , we write “YES” for a blue pair if the two-separators rule holds (so that pair produces one crossing in its $K_{2,2}$), and “NO” otherwise. Let $\tau(p, q)$ be the number of YES’s among the three blue pairs when $q \in W$.

W1 (middle): wedge outside the side between pb_1 and pb_2 . Here the line $b_1 b_2$ separates p and q , while $b_1 b_3$ and $b_2 b_3$ do not. Also, the red line pq separates b_1 and b_2 , but not the third point from either. Hence:

$$\{b_1, b_2\} : \text{YES}, \quad \{b_2, b_3\} : \text{NO}, \quad \{b_1, b_3\} : \text{NO}.$$

So $\tau(p, q) = 1$.

W2 (middle): wedge outside the side between pb_2 and pb_3 . By the same reasoning with indices shifted,

$$\{b_2, b_3\} : \text{YES}, \quad \{b_1, b_2\} : \text{NO}, \quad \{b_1, b_3\} : \text{NO}.$$

So $\tau(p, q) = 1$.

W3 (middle): wedge outside the side between pb_3 and pb_1 . Again by symmetry,

$$\{b_1, b_3\} : \text{YES}, \quad \{b_1, b_2\} : \text{NO}, \quad \{b_2, b_3\} : \text{NO}.$$

So $\tau(p, q) = 1$.

W4 (flank): wedge adjacent to pb_1 on the *outside*. Moving from p to q crosses the extensions of all three sides, so every blue line $b_i b_j$ separates p and q . The red line pq also separates each blue pair: it separates b_1 from b_2 , separates b_1 from b_3 , and because q is on the outside flank it also cuts across the base and separates b_2 from b_3 . Hence

$$\{b_1, b_2\} : \text{YES}, \quad \{b_2, b_3\} : \text{YES}, \quad \{b_1, b_3\} : \text{YES},$$

so $\tau(p, q) = 3$.

W5 (flank): wedge adjacent to pb_3 on the *outside*. Mirror of W4. All three blue pairs pass the two-separators rule, so $\tau(p, q) = 3$.

W6 (middle): the remaining outward wedge (opposite one of W1–W3). Same pattern as its opposite middle wedge: exactly the facing blue pair passes the rule. So $\tau(p, q) = 1$.

These six mini-proofs match the labels in the figure:

$$\underbrace{1, 1, 1, 1}_{\text{four middle wedges}}, \quad \underbrace{3, 3}_{\text{two flank wedges}}.$$

Add up over all red pairs

Let R be the set of red points and write N_F for the number of unordered red pairs $\{p, q\}$ with q in a *flank* wedge of p (labels 3), and $N_M = \binom{n}{2} - N_F$ for the number in *middle* wedges (labels 1).

Lemma 4 (Pairwise sum). *Every crossing belongs to exactly one unordered red pair. Therefore*

$$\text{cr} = \sum_{\{p, q\} \subset R} \tau(p, q) = 3N_F + 1N_M.$$

Proof. Each crossing occurs in exactly one small $K_{2,2}$ formed by a red pair and a blue pair. It is counted in the $\tau(p, q)$ for that red pair and nowhere else. $\square$

Theorem 5 (Main formula and bounds). *For any placement of the n red points,*

$$\boxed{\text{cr} = \binom{n}{2} + 2N_F}, \quad \binom{n}{2} \leq \text{cr} \leq 3\binom{n}{2}.$$

The lower end is achieved by placing all red points on a very thin ray so every pair is middle. The upper end (within this classifier) is achieved by clustering all red points tightly in a flank wedge so every pair is flank.